Prodigy Anesthesia Quick Review

Inhalation Anesthetics

According to the Occupational Safety and Health Administration (OSHA), operating room personnel should not be exposed to more than how many parts per million of a halogenated agent? What if nitrous oxide is used?

If nitrous oxide is also used, the maximum exposure to halogenated agents allowed by OSHA is 0.5 ppm. If nitrous oxide is not used, then the maximum exposure is 2 ppm. The maximum exposure to nitrous alone is 25 ppm.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 280.

What is MAC?

MAC represents "the Mean Alveolar Concentration that prevents movement in half of subjects in response to a surgical incision."

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.

What is MAC-BAR?

MAC-BAR refers to the concentration of inhaled anesthetic that prevents an adrenergic response to skin incision (increased heart rate, blood pressure, plasma norepinephrine levels). MAC-BAR50 is the dose of volatile anesthetic that prevents the adrenergic response in 50% of patients and MAC-BAR95 is the dose that prevents an adrenergic response in 95% of patients.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 88.

What is MAC-awake and how does it correlate to MAC?

Mac-awake is defined as "the concentration of inhaled anesthetic that inhibits appropriate responses to spoken commands in half of patients". It normally correlates to approximately 0.35 MAC.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.

How does having red hair affect MAC?

It has been demonstrated that natural redheads have a significantly higher anesthetic requirement for desflurane than do non-redhead patients.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.

MAC and MAC-awake are reduced by neuraxial blockade. How does this occur?

Even though neuraxial blockade doesn't affect cranial nerve function like other factors that decrease MAC (hypothermia, hypnotics, opioids, antidepressants, anti-seizure medications, etc) it decreases MAC by inhibiting the ascending spinal cord signals that stimulate cortical arousal in the brainstem.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607-608.

What is ED95 and how does it relate to MAC?

The ED95 is the dose of a volatile anesthetic that is effective in preventing movement in response to surgical incision in 95% of patients and usually correlates to about 1.3 MAC.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.

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How does body temperature affect MAC?	For each 1 degree Celsius drop in temperature, MAC decreases by 5%. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.*
Besides hypothermia and advanced age, what other physiologic factors decrease MAC?	Besides hypothermia and advanced age, factors that decrease MAC include alpha-2 agonists, acute ethanol ingestion, hypoxemia, hyponatremia, metabolic acidosis, anemia, hypotension, and pregnancy. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.*
How does age affect MAC and at what age is MAC highest?	MAC generally decreases with age. MAC is highest in patients between the ages of 6 and 12 months. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 607.*
What are the four groups tissues are divided into when discussing the transfer of volatile anesthetics to the tissues?	The four groups are the vessel-rich group, muscle, fat, and the vessel-poor group. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
What tissues constitute the vessel-rich group, what percentage of body weight do they account for, and what percent of cardiac output do they receive?	The vessel-rich group consists of the brain, heart, liver, kidneys, and endocrine organs. They constitute about 10% of body weight, but receive about 75% of cardiac output. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
What tissues constitute the muscle group, what percentage of body weight do they account for, and what percent of cardiac output do they receive?	The muscle group consists of both muscle and skin. It is not as well perfused as the vessel-rich group, comprising about 50% of body weight but receiving only 19% of cardiac output. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
What tissues constitute the fat group, what percentage of body weight do they account for, and what percent of cardiac output do they receive?	Adipose tissue is the sole component of the fat group. It usually is about 20% of bodyweight and receives 6% of the cardiac output. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
What tissues constitute the vessel-poor group, what percentage of body weight do they account for, and what percent of cardiac output do they receive?	The vessel-poor group consists of teeth, hair, nails, bones, ligaments, and cartilage. It comprises about 20% of body weight and receives less than 1% of cardiac output. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*

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What is the Meyer-Overton rule?	The Meyer-Overton rule states that the lipid solubility of an inhalation agent is directly proportional to its potency. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 86.*
What is Dalton's law and what is meant by partial pressure?	Partial pressure refers to the fractional pressure a gas exerts as a part of all of the gases with which it is mixed. Dalton's law states that the total pressure of a mixture of gases is equal to the sum of the pressures each individual gas in the mixture exerts. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 449-450.*
What are the three factors that influence the uptake of a volatile anesthetic?	Alveolar blood flow, the blood:gas solubility of the anesthetic, and the difference between the partial pressure of the anesthetic in the alveoli and in venous blood. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
What is meant by an FA/FI ratio of 1.0?	The degree to which the alveolar concentration of an anesthetic is becoming closer to the inspired concentration of an anesthetic is often expressed as a ratio. If the alveolar concentration equals the inspired concentration, then their ratio = 1:1 or 1.0. Stating 'the rate at which FA/FI = 1' is just a shortcut for stating 'the rate at which the alveolar concentration of the anesthetic equals the inspired concentration of the anesthetic'. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 155-156.*
How does an increase in the uptake of a volatile anesthetic affect the rate of induction?	The greater the rate of uptake, the greater the difference between the alveolar concentration (FA) and the inspired concentration (FI) and the slower the rate of induction. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 155.*
Besides the vaporizer dial setting, upon what three factors does the composition of the inspired gas mixture a patient receives depend?	The composition of the inspired gas mixture a patient receives depends primarily upon the vaporizer dial setting, fresh gas flow rate, the volume of the breathing circuit, and any absorption by the circuit. The inspired gas concentration will be closer to the fresh gas concentration if the circuit volume and level of absorption by the circuit are low and the fresh gas flow is high. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 155.*

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How is the uptake of volatile anesthetics related to the rate of rise of the FA/FI ratio?

If there was no uptake of anesthetic gas by the body, the concentration of anesthetic gas in the alveoli (FA) would rapidly equal the concentration of anesthetic gas in the inspired gas (FI). FA lags behind FI because the pulmonary circuit takes up anesthetic gas and lowers the concentration in the alveoli. The greater the amount of uptake, the slower the FA/FI ratio rises toward 1. By definition then, uptake is inversely related to the rate of rise of the FA/FI ratio.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 155.

How does an increase in minute ventilation affect the rate at which FA approaches FI?

As the minute alveolar ventilation increases, the rate at which the alveolar concentration of a volatile anesthetic approaches the inspired concentration increases. They are directly proportional.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 596.

How does low blood solubility affect the pulmonary uptake and elimination of a volatile anesthetic?

The lower the blood solubility, the faster pulmonary uptake and elimination occurs.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 596.

What is overpressurization of a volatile anesthetic?

Overpressurization is the administration of 2-3 times the desired alveolar concentration of a gas until the desired depth of anesthesia is reached. At that point, the concentration of inspired gas is reduced to a lower level.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 601.

What are the factors that determine the rate at which the alveolar concentration (FA) of an anesthetic approaches the inspired concentration (FI)?

The rate at which FA approaches FI depends upon minute ventilation, cardiac output, and the blood:gas coefficient of the anesthetic.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 596.

How does a right to left shunt affect the speed of induction?

If a right to left shunt is present, an inhalation induction proceeds more slowly because the anesthetic concentration of arterial blood increases more slowly. The opposite is true with a left to right shunt because the rate of transfer of anesthetic from the lungs to the blood is more rapid. This effect is rarely evident in the clinical setting, however.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 159.

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Why is the alveolar partial pressure of an anesthetic agent important?	The alveolar partial pressure is important because it is the factor that determines the partial pressure of anesthetic gas within the blood. In turn, the partial pressure of anesthetic gas in the blood determines the partial pressure of anesthetic gas within the brain. Thus, the partial pressure of anesthetic gas in the alveoli determines the concentration of gas within the brain and this is the determinant of the clinical level of anesthesia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156.*
Which agent would have a faster onset at induction, a highly soluble anesthetic agent or a poorly soluble agent?	A poorly soluble agent would not be absorbed from the alveoli into the blood as quickly, therefore, its alveolar partial pressure would rise more quickly. Because the alveolar partial pressure determines the partial pressure of the gas in the brain where it exerts its clinical effect, the poorly soluble agent would have a faster onset than a highly soluble agent. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156-157.*
Anesthetic A has an blood:gas partition coefficient of 0.5 and anesthetic B has a blood:gas coefficient of 0.75. All other factors being equal, which one will lead to a slower induction and why?	Anesthetic B will exhibit a slower induction. The higher the blood:gas coefficient, the greater the uptake of the anesthetic and the longer it will take for the FA/FI to equal 1.0 *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 157.*
How do changes in cardiac output affect the rate at which the alveolar concentration of a volatile anesthetic approaches the inspired concentration?	They are inversely proportional. As cardiac output increases, the rate at which FA approaches FI decreases and vice versa. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 596.*
How does the blood:gas coefficient affect the rate at which the alveolar concentration of a volatile anesthetic approaches the inspired concentration?	They are inversely proportional. The higher the blood:gas solubility coefficient, the slower the rate at which FA approaches FI. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 596.*

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How does the rate of rise in the FA/FI ratio compare between soluble and insoluble anesthetic gases?	Anesthetic gases with low blood:gas solubility coefficients are not readily taken up by the bloodstream. Consequently, the alveolar partial pressure of these agents does not drop rapidly. Because the alveolar partial pressure does not drop due to the gas being taken up, it rapidly approaches the inspired concentration of the gas. Conversely, highly soluble agents are absorbed rapidly from the alveoli into the blood and the alveolar pressure drops substantially. Because of this absorption, the alveolar concentration rises much more slowly and takes a longer time to reach the inspired concentration. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 156-157.*
What is the primary way in which volatile anesthetics are eliminated from the body?	Although metabolism accounts for a very small amount of the elimination, the vast majority is eliminated via the alveoli. The gases are exhaled unchanged. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 159.*
To what extent are desflurane, isoflurane, and sevoflurane metabolized?	Remember the rule of Two's: Desflurane = 0.02%, isoflurane = 0.2%, and sevoflurane = 2% *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 598.*
What is the primary way in which all volatile anesthetics are metabolized?	The biodegradation of all volatile anesthetics is via phase I oxidation in the liver. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 88.*
How do the volatile anesthetics affect cardiac conduction?	Sevoflurane, isoflurane, and desflurane all prolong the QT interval. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 612.*
How do volatile anesthetics affect hepatic blood flow?	All volatile anesthetics reduce hepatic blood flow primarily by decreasing portal blood flow. Isoflurane decreases it the least because it allows for compensatory increases in flow via the hepatic artery. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 613.*

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How do volatile anesthetics affect pulmonary vascular resistance?	While the vascular smooth muscle throughout the body is decreased, the effects on the pulmonary vasculature is minimal. The combination of changes in cardiac output counteract the vasodilatory actions of the volatile anesthetics resulting in no significant changes in pulmonary artery pressures. Nitrous oxide is capable of increasing pulmonary vascular resistance to a small degree, but this is not considered significant except in patients with pre-existing pulmonary hypertension. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 469.*
What is hypoxic pulmonary vasoconstriction and how do volatile anesthetics affect it?	Although volatile anesthetics exert little effect on pulmonary blood flow, they do exert a significant effect on hypoxic pulmonary vasoconstriction, which is the localized vasoconstriction that occurs in response to regional hypoxia in the lung. It occurs due to alveolar hypoxia, not arterial hypoxemia. HPV normally serves to improve V/Q characteristics within the lung to maximize the interface between oxygen and pulmonary blood flow. All of the volatile anesthetics have been shown to attenuate HPV in high doses in animal models, but the effect has not been demonstrated as clearly in clinical doses in humans yet. HPV can also be inhibited by very high or very low pulmonary artery pressures, hypocapnia, beta-adrenergic agonists, vasodilators such as nitroglycerin and nitroprusside, pulmonary infection, high or very low mixed venous PO2, and calcium channel blockers. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 674-676.*
How can volatile anesthetic agents be used to treat status asthmaticus?	Because they exert a powerful bronchodilating characteristic, induction of general anesthesia with high concentrations of volatile anesthetics has been used in the treatment of refractory status asthmaticus. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 94-95.*
How does the administration of a volatile anesthetic affect renal perfusion and the glomerular fitration rate?	Autoregulation of blood flow through the kidneys remains largely intact when volatile anesthetics are used. As the mean arterial pressure decreases from the effects of volatile anesthetics, the renal vasculature dilates to increase blood flow. This compensatory reduction in renal vascular resistance can, however, lead to a decrease in the glomerular filtration rate. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 95.*
Which of the inhalation anesthetics trigger malignant hyperthermia?	All of the volatile agents trigger malignant hyperthermia. Nitrous oxide is considered at most to be a weak trigger of malignant hyperthermia and is often combined with an intravenous anesthetic for patients with malignant hyperthermia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 96.*

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Which volatile anesthetic decreases CMRO2 the	Isoflurane decreases CMRO2 the most. The effects of sevoflurane and desflurane are roughly equivalent. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 582.*
Which volatile agent increases cerebrospinal fluid absorption?	Isoflurane is unique in that it is the only volatile agent that facilitates CSF absorption and has a favorable effect on CSF dynamics. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 581.*
In reference to the volatile anesthetics, what is luxury perfusion?	The use of volatile anesthetics can result in luxury perfusion, a potentially beneficial effect during global ischemia which is characterized by a decrease in CMRO2 with a simultaneous increase in cerebral blood flow. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 582.*
What is the only volatile agent capable of producing an isoelectric EEG at clinical doses (1-2 MAC)?	Isoflurane is the only volatile agent capable of producing an isoelectric EEG at clinical doses. Desflurane and sevoflurane can produce burst suppression at doses greater than 1.2 and 1.5 MAC respectively, but not electrical silence. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 169.*
What inhalation anesthetic can increase the intracranial pressure substantially?	Nitrous oxide can increase intracranial pressure substantially, especially in patients with mass lesions. Desflurane can also increase intracranial pressure in a small but steady manner, probably due to altered cerebrospinal fluid dynamics. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 878.*
How do inhalation anesthetics affect tidal volume and respiratory rate?	The tidal volume is decreased as the concentration of the agent increases. The respiratory rate increases, however, and this is typically sufficient to prevent increases in arterial CO2 due to hypoventilation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 94.*
How do volatile anesthetics affect the respiratory response to carbon dioxide?	The response to CO2 is diminished. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 94.*

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How do volatile anesthetics affect heart rate?	All volatile anesthetics increase heart rate by a direct vagolytic effect on the heart and via a baroreceptor response to the decrease in blood pressure they produce. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 612.*
How do volatile anesthetics affect mean arterial pressure?	All volatile anesthetics decrease the MAP in a dose-dependent fashion by decreasing vascular resistance, increasing vascular compliance, and depressing the myocardium. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 611.*
How do inhaled anesthetic gases affect cerebral blood flow and intracranial pressure?	All the inhaled anesthetic gases decrease cerebrovascular resistance, increase cerebral blood flow and, as a result, increase intracranial pressure. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 165.*
How do inhalation agents affect myocardial contractility?	Desflurane, isoflurane, and sevoflurane reduce myocardial contractility, delay the relaxation of the cardiac chambers, and can impair the synchronization between the left atrial and left ventricular contractions. Nitrous oxide reduces myocardial contractility by decreasing cardiac myocyte calcium levels. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 612.*
How do inhalation anesthetics affect the hypoxic ventilatory drive? In what situations might this be significant?	Concentrations as little as 0.1 MAC of a volatile anesthetic can abolish the body's ventilatory response to hypoxia. This may become significant in patients whose minute ventilation is largely determined by their hypoxic drive, such as patients with emphysema. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 94.*
How do volatile anesthetics affect cerebral metabolism?	The cerebral metabolic rate of oxygen consumption (CMRO2) is decreased by volatile anesthetics in a dose-dependent fashion. The exception in anesthetic gases is nitrous oxide, which can increase the CMRO2. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 89.*
How do volatile anesthetics affect neuromuscular function?	All volatile agents produce a dose-dependent relaxation of skeletal muscle and a potentiation of the effects of non-depolarizing muscle relaxants. This is predominantly due to effects at the postsynaptic neuromuscular junction membrane. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 96.*
What is the vapor pressure of isoflurane?	238 *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 597.*

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What is the vapor pressure of desflurane?	664 *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 597.*
What is the vapor pressure of sevoflurane?	157 *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 597.*
What are the blood:gas partition coefficients of isoflurane, desflurane, sevoflurane, and nitrous oxide?	Isoflurane = 1.43, desflurane = 0.42, sevoflurane = 0.68, and nitrous oxide = 0.47 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 86.*
How many fluorides do desflurane, isoflurane, and sevoflurane each possess?	Isoflurane has 5 fluorides, desflurane has 6, and sevoflurane is the most fluorinated with 7 fluoride atoms. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 597.*
What are the oil:gas partition coefficients for isoflurane, desflurane, sevoflurane, and nitrous oxide?	Isoflurane = 99, desflurane = 19, sevoflurane = 50, and nitrous oxide = 1.4 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 86.*
What are the molecular weights of nitrous oxide, isoflurane, desflurane, and sevoflurane?	Nitrous oxide is 44 daltons, isoflurane is 184.5, desflurane is 168, and sevoflurane is 200.1. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 597.*
What is the MAC of sevoflurane in a healthy, 30-60 year old with a normal body temperature?	2.0 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 88.*
What is the MAC of desflurane in a healthy, 30-60 year old with a normal body temperature?	6 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 88.*
What is the MAC of isoflurane in a healthy, 30-60 year old with a normal body temperature.	1.17 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 88.*
Between desflurane, isoflurane, sevoflurane, and nitrous oxide, which inhaled anesthetic is the most potent?	Isoflurane is the most potent. The oil:gas solubility coefficient provides an indication to the potency of an inhalation anesthetic. The higher the oil:gas solubility, the greater the potency. The oil:gas partition coefficient of isoflurane is 99, for sevoflurane it is 50, for desflurane it is 18.7, and for nitrous oxide it is 1.4. *Nagelhout JJ, Plaus K. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 94.*

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How can desflurane affect the heart rate in high concentrations?

Desflurane produces a small increase in sympathetic activity. When the concentration of desflurane is increased rapidly, catecholamine release is more pronounced and can result in increases in heart rate and blood pressure.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 170.

Comparing desflurane, sevoflurane, nitrous oxide and isoflurane, an increase in cardiac output would slow the onset of which inhalation anesthetic the most?

Isoflurane. The effect is greatest in agents with a high blood:gas partition coefficient. The blood:gas partition coefficient of desflurane is 0.42, for nitrous oxide it is 0.47, for sevoflurane it is 0.6, and for isoflurane it is 1.4.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 81.

What is compound A and what volatile anesthetic is associated with it?

Compound A is a vinyl ether that is formed from the degradation of sevoflurane in carbon dioxide absorbents. It has been linked to renal damage in animal studies. No adverse effects have been demonstrated in humans at clinical doses, but the general recommendation is to use at least 2L/min fresh gas flow rate when using sevoflurane as a dilutional measure to prevent the accumulation of compound A in any significant concentration.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 172.

What is the most common arrhythmias seen with sevoflurane? With nitrous oxide?

The most common arrhythmias seen with the use of sevoflurane are isolated supraventricular ectopic beats. Nitrous can produce atrioventricular nodal rhythms.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 612.

Isoflurane has been demonstrated to cause coronary steal in some cases. What is meant by this?

Isoflurane dilates coronary arteries. Diseased vessels with fixed, stenotic lesions, however cannot dilate as effectively as normal vessels. As the normal vessels dilate, blood is shunted towards the normal areas of the heart and away from the areas suplied by the already diseased vessels. This can result in worsening of ischemia. Desflurane and sevoflurane do not cause this effect.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 612.

Why is desflurane not used for inhalation inductions in pediatric patients?

Compared to sevoflurane and nitrous oxide, desflurane acts as a respiratory irritant in concentrations above 6% and can result in breath-holding, coughing, and laryngospasm during induction.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 94.

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Despite having a lower blood:gas solubility coefficient, the rate of rise of the FA/Fi ratio is higher for nitrous oxide than for desflurane. Why is this?

Agents with a lower blood:gas partition coefficient exhibit a faster rate of rise in the FA/Fi ratio. The exception is that nitrous oxide will exhibit a faster rate of rise than desflurane despite the fact that it has a blood:gas partition coefficient of 0.47 and desflurane has a blood:gas partition coefficient of 0.42. The increase with nitrous oxide is due to the fact that it is administered in doses of 50-70% compared to 3-9% for desflurane. This is referred to as the concentration effect.

Nagelhout JJ, Plaus K. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 94-95.

Nitrous oxide is 35 times more soluble in blood than nitrogen. What are the implications of this?

Because nitrous oxide is so much more soluble that nitrogen, it can diffuse into an air-filled space faster than the nitrogen can diffuse out of it, resulting in increased pressure, volume, or both in the space it enters. If the air-filled space is in the stomach or intestines, it can result in distention. If it is a pneumothorax or air embolus, the effects can be disastrous. Nitrous is even capable of diffusing into an endotracheal tube balloon, increasing the pressure it exerts against the tracheal wall.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 166.

What is diffusion hypoxia?

The alveolar elimination of nitrous oxide occurs at such a rapid rate that alveolar oxygen and carbon dioxide are diluted. As a result, alveolar oxygen levels can drop below 21%. The administration of 100% oxygen after the discontinuation of nitrous oxide prevents this from occurring.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 159.

What is methionine synthetase and how does nitrous oxide affect it?

Methionine synthetase is a cobalamin-dependent enzyme necessary for nerve myelination, homocysteine degradation, and DNA synthesis. Nitrous oxide oxidizes the cobalt atom found on vitamin B12 (cobalamin) which can inhibit the formation of methionine synthetase. This would exert little effect on normal, healthy patients, but may warrant consideration of the use of nitrous oxide in patients with severe B12 deficiency.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 613.

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